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Indian Standard

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(Reaffirmed 2021)

मोबाइल फोन हैंडसेट

भाग 1 सुरक्षा अपेक्षाएँ

Mobile Phone Handsets

Part 1 Safety Requirements

ICS 33.050.10

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Price Group 3

Information and Communication Technology Sectional Committee, LITD 13

FOREWORD

This Indian Standard (Part 1) was adopted by the Bureau of Indian Standards, after the draft finalized by the Information and Communication Technology Sectional Committee had been approved by the Electronics and Information Technology Division Council.

Mobile phone handset is being used by the vast cross-section of consumers across all segments of people in the country. It is also reported that inconsistent quality of product being available in the open market may affect interest of common consumers.

There has been demand from users of the device and also from various Ministries to develop Indian Standard on mobile phone handset. Accordingly an effort has been made to prepare this document. This standard has been formulated in two parts. The other part of the standards in the series is as follows:

Part 2 Performance requirements

While formulating this standard it has been found that there is no single comprehensive specification available at the international level. However, various requirements of the product have been specified separately. In India basic regulatory/interoperability requirements have been specified by Government of India agencies like TRAI/DOT. To cater to the need expressed from various quarters, a comprehensive document has been prepared which can be considered for implementation by the concerned agencies.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

MOBILE PHONE HANDSETS

PART 1 SAFETY REQUIREMENTS

1 SCOPE

This standard (Part 1) specifies the safety requirements for mobile phone handsets.

2 REFERENCES

The standards listed in Annex A contain provisions, which through reference in this text constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated in Annex A.

3 TERMS AND DEFINITIONS

For the purpose of this standard, the following definitions shall apply. Additional terms and definitions as given in Annex B may also be referred.

3.1 Mobile Phone Handset — An wireless electronic device used for communication both voice and data using different technologies.

3.2 International Mobile Station Equipment Identity (IMEI) — An IMEI is a serial number that uniquely identifies a GSM or UMTS mobile phone. Typically 15 digits long, the IMEI code is broken into sections that provide information about a phone, such as its manufacturer, to the mobile network that the phone is connected to.

3.3 MEID — An MEID is a unique identifier found on CDMA handsets. The MEID system is compatible with the existing IMEI number system used in GSM and UMTS handsets. It is being phased in to replace the ESN numbers currently used on CDMA devices, since the pool of ESN numbers has been virtually exhausted due to the number of devices in use.

3.4 SIM — A SIM card is a small card that contains a mobile network subscriber's account information. This allows the phone using the card to attach to a mobile network.

3.6 Specific Absorption Rate (SAR) — The time derivative of the incremental electromagnetic energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of given mass density (ρ)

$$SAR = \frac{d}{dt} \left(\frac{dW}{dm} \right) = \frac{d}{dt} \left(\frac{dW}{\rho dV} \right)$$

SAR can be obtained using either of the following equations:

$$SAR = \frac{\sigma_E^2}{\rho}$$

$$SAR = c_h \left. \frac{dT}{dt} \right|_{t=0}$$

where

SAR = specific absorption rate, W/kg;

E = r.m.s value of the electric field strength in the tissue, V/m;

σ = conductivity of the tissue, Siemens/m;

ρ = density of the tissue, kg/m³;

c_h = heat capacity of the tissue, J/kg/K; and

$\left. \frac{dT}{dt} \right|_{t=0}$ = initial time derivative of temperature in the tissue, K/s.

4 REQUIREMENTS

4.1 Operating Frequency

Mobile phone handsets shall be capable of operating at the frequency bands as notified by DOT from time to time.

4.2 Safety Requirements

Mobile phone handset shall meet applicable requirements of IS 13252 (Part 1). In addition the following shall apply:

- a) Mobile phone handsets shall comply with the SAR value of 1.6 W/kg averaged over 1gm tissue-SAR.
- b) Mobile phone handsets shall display the max value of SAR by way of request using MMI string code *#07# as per the 3GPP specification.
- c) The SAR (Head) measurement shall be carried out in accordance with IS/IEC 62209-1.

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4.3 Restriction on Use of Hazardous Material (RoHS)

Mobile phone handsets shall not contain lead, mercury, cadmium, hexavalent chromium, poly brominated biphenyls or poly brominated diphenyl ethers as per relevant guidelines provided by Ministry of Environment and Forest. The requirement shall be verified in accordance with IS 16197 (All relevant parts).

4.4 Mobile Phone Battery

Batteries used in the mobile phone handsets shall conform to IS 16046 and IS 16047.

4.5 EMI/EMC Requirements

The mobile phone handset shall meet the EMC requirement as applicable as per following:

Requirement	Application	Applicable Standard
EMC	General	EN 301 489-1
	NFC	EN 301 489-3
	2G	EN 301 489-7
	BT, Wi-Fi	EN 301 489-17
	GPS	EN 301 489-19
	3G, LTE	EN 301 489-24

4.6 RF Requirement

The mobile phone handset shall conform to the following as applicable:

Requirement	Application	Applicable Standard
Radio	2G RF, RSE	EN 301 511
	3G RSE	EN 301 908-1
	3G RF	EN 301 908-2
	LTE RF	EN 301 908-13
	BT, Wi-Fi Radio	EN 300 328
	Wi-Fi Radio 802.11 b/g/n 802/11/a/n/ac	EN 301 893

4.7 Protocol Conformance

Mobile phone handset should conform to the following protocol standards as applicable:

- a) GSM/EGPRS — Protocol Conformance :
3GPP TS 51.010-1 : Digital cellular telecommunications system (Phase 2+) Mobile Station (MS) Conformance Specification; Part 1 Conformance specification

- b) WCDMA/UMTA (Multimode) – Protocol Conformance:

- 1) TS 34.121 — Terminal conformance specification, Radio transmission and reception (FDD).
- 2) TS 34.123 — User Equipment (UE) conformance specification; Part 1: Protocol conformance specification, subset.
- 3) TS 34.124 — Electromagnetic compatibility (EMC) requirements for mobile terminals and ancillary equipment.
- 4) TS 31.121 — UICC terminal interface; Universal Subscriber Identity Module (USIM) application test specification.
- 5) TS 34.171 — Terminal conformance specification; Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)(Applicable only, if handset support A-GPS).

4.8 International Mobile Station Equipment Identity (IMEI)

- a) Each individual Mobile phone handset shall be allocated a unique 'International Mobile Station Equipment Identity (IMEI)'. The IMEI number of the mobile phone handset shall be available in the latest updated IMEI database of the GSMA. The mobile phone handset IMEI shall not be with all zeroes. For the case of CDMA the Mobile phone handset shall have a valid and unique Electronic Serial Number (ESN) or Mobile Equipment Identifier (MEID) number :
- b) Mobile phone handset shall display all the IMEI no. (GSM/UMTS/LTE) or MEID/ ESN(CDMA) by way of request using the MMI string code *#06# as per 3GPP requirements.
- c) In case mobile phone handset having more than one SIM, if each SIM is associated with its own transceiver, then each transceiver/SIM slot shall have its own associated IMEI number. Therefore, a dual SIM Mobile phone handset shall have 2 IMEI numbers; a three SIM mobile phone handset shall have 3 IMEI numbers, etc.
- d) In case of dual transceiver (Dual technology, GSM and CDMA) mobile phone handset with single SIM device, the mobile phone handset shall display two unique IMEI and MEID/ ESN numbers. Each additional SIM slot/ transceiver shall add one unique IMEI/MEID/ ESN number to be displayed

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5 MARKING

5.1 The mobile phone handset shall be indelibly and clearly marked with the manufacturer's brand identification mark and model.

5.2 Each mobile phone handset should be packed with an instruction manual to furnish the following information:

- Name or trade-mark of the manufacturer;
- Precaution to be taken in the use of hearing aid; and
- Any other useful information the manufacturer would like to furnish.

5.3 BIS Certification Marking

The mobile phone handset may also be marked with the Standard Mark.

5.3.1 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and the Rules and Regulations made thereunder.

The details of conditions under which a license for the use of the Standard Mark may be granted to the manufacturers or producers may be obtained from the Bureau of Indian Standards.

6 TESTS

The following tests shall be considered as type tests:

- General safety requirements as per IS 13252 (Part 1) as applicable,
- Operating frequency (*see 4.1*),
- SAR value measurement (*see 4.2*),
- RoHS compliance (*see 4.3*),
- Mobile phone battery requirement (*see 4.4*),
- EMI/EMC requirements (*see 4.5*),
- RF requirement (*see 4.6*),
- Protocol conformance (*see 4.7*), and
- International mobile station equipment identity (IMEI) requirement (*see 4.8*).

ANNEX A

(Clause 2)

LIST OF REFERRED INDIAN STANDARDS

IS No./ International Standard	Title	IS No./ International Standard	Title
13252(Part 1) : 2010	Information technology equipment — Safety: Part 1 General requirements (<i>second revision</i>)		models, inst. and procedures-Part 1: Procedure to determine the specific absorption rate(SAR)for hand-held devices used in close proximity to the ear(freq. range of 300 MHZ to 3 GHZ)
16046 : 2012	Secondary cells and batteries containing alkaline or other nonacid electrolyte safety requirement for portable sealed secondary cells and for batteries made from them, for use in portable applications	EN 300 328	Electromagnetic compatibility and Radio spectrum Matters (ERM); Wideband transmission systems; Data transmission equipment operating in the 2,4 GHz ISM band and using wide band modulation techniques; Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive
16047 : 2012	Secondary cells and batteries containing alkaline or other non-acid electrolytes - Secondary lithium cells and batteries for portable applications		Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility
16197 (All relevant parts)	Determination of certain substances in electrotechnical products		
IS/IEC 62209-1 : 2012	Human exposure to radio freq. fields from hand-held and body-mounted wireless comm. devices-Human	EN 301 489-1	

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<i>IS No./ International Standard</i>	<i>Title</i>	<i>IS No./ International Standard</i>	<i>Title</i>
	(EMC) standard for radio equipment and services; Part 1: Common technical requirements		in the GSM 900 and GSM 1800 bands covering essential requirements of article 3.2 of the R&TTE directive (1999/5/EC)
EN 301 489-3	Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 3: Specific conditions for Short-Range Devices (SRD) operating on frequencies between 9 kHz and 40 GHz	EN 301 893	Broadband Radio Access Networks (BRAN); 5 GHz high performance RLAN; Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive
EN 301 489-7	Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 7: Specific conditions for mobile and portable radio and ancillary equipment of digital cellular radio telecommunications systems (GSM and DCS)	EN 301 908-1	IMT cellular networks; Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive; Part 1: Introduction and common requirements
EN 301 489-17	Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility (EMC) standard for radio equipment; Part 17: Specific conditions for Broadband Data Transmission Systems	EN 301 908-2	IMT cellular networks; Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive; Part 2: CDMA Direct Spread (UTRA FDD) User Equipment (UE)
EN 301 489-19	Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 19: Specific conditions for Receive Only Mobile Earth Stations (ROMES) operating in the 1,5 GHz band providing data communications	EN 301 908-13	IMT cellular networks; Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive; Part 13: Evolved Universal Terrestrial Radio Access (E-UTRA) User Equipment (UE)
EN 301 489-24	Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 24: Specific conditions for IMT-2000 CDMA Direct Spread (UTRA and E-UTRA) for Mobile and portable (UE) radio and ancillary equipment	3GPP TS 31.121	UICC-terminal interface; Universal Subscriber Identity Module (USIM) application test specification
EN 301 511	Global System for Mobile communications (GSM); Harmonized EN for mobile stations	3GPP TS 34.121	Terminal conformance specification, Radio transmission and reception (FDD)
		3GPP TS 34.123	User Equipment (UE) conformance specification; Part 1: Protocol conformance specification, subset
		3GPP TS 34.124	Electromagnetic compatibility (EMC) requirements for mobile terminals and ancillary equipment
		3GPP TS 34.171	Terminal conformance specification; Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)-(Applicable only if handset support A-GPS)
		3GPP TS 51.010-1	Digital cellular telecommunications system (Phase 2+) Mobile Station (MS) Conformance Specification; Part 1 Conformance specification

ANNEX B

(Clause 3)

ADDITIONAL TERMS AND DEFINITION

C-1 GSM (Global System for Mobile Communications) — Originally Groupe Spécial Mobile, is a standard developed to describe protocols for second generation (2G) digital cellular networks used by mobile phones.

C-2 CDMA — Code division multiple access (CDMA) describes a communication channel access principle that employs spread-spectrum technology and a special coding scheme (where each transmitter is assigned a code). By contrast, time division multiple access (TDMA) divides access by time, while frequency-division multiple access (FDMA) divides it by frequency. CDMA is a form of ‘spread-spectrum’ signalling, since the modulated coded signal has a much higher bandwidth than the data being communicated.

C-3 WCDMA — WCDMA stands for Wideband Code Division Multiple Access, and is the 3G technology, as defined by the ITU’s IMT-2000 specification, that is used by UMTS and NTT DoCoMo’s FOMA network. WCDMA makes use of many core CDMA technologies created by Qualcomm, though not nearly as many as do regular CDMA carrier networks. WCDMA networks and devices are not compatible with regular CDMA networks and devices, they merely share some of the same core technologies.

C-4 ESN — An ESN is a numeric identifier that uniquely identifies a CDMA phone. The ESN can typically be found written underneath a phone’s battery

and is generally written in both decimal and hexadecimal versions. The ESN is what a CDMA network uses to identify a phone and determine which subscriber’s account, if any, it is linked to. Because of this, when switching from one phone to another, subscribers will have to provide the ESN of the new phone to the network carrier before it can be activated.

C-5 UMTS — UMTS, short for Universal Mobile Telecommunications System, is a 3G networking standard used throughout much of the world as an upgrade to existing GSM mobile networks. UMTS makes use of WCDMA, a technology that shares much with CDMA networks used throughout the world, though it is not compatible with them. Base level UMTS networks are generally capable of downlink speeds as fast as 384kbps. Newer HSDPA variants are capable of rates as high as 3.6Mbps or more. Originally used only on the 2100MHz frequency band in Europe, UMTS is now supported on the 850MHz and 1900MHz bands in North America.

C-6 LTE — LTE, short for Long Term Evolution, is considered by many to be the obvious successor to the current generation of UMTS 3G technology, which is based upon WCDMA, HSDPA, HSUPA, and HSPA. LTE is not a replacement for UMTS in the way that UMTS was a replacement for GSM, but rather an update to the UMTS technology that will enable it to provide significantly faster data rates for both uploading and downloading.

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

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Amendments Issued Since Publication

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